

ISM 6251 and 4543: Machine Learning (Fall 2025)

Assignment 2

- Please provide succinct answers to the questions below.
- You should submit a .ipynb file and a pdf (for any non-coding questions) in Canvas.
- Please include the names of all team-members in your write up and in the name of the file. One submission per group.
- You should include all the Python functions you use.

Problem 1. The dataset *income_data.csv* represents an extract from a commercial marketing database. You can find the column description in *income_info.txt*. The goal is to fit a decision tree to predict the annual income of a household from 13 demographic attributes and interpret the results. We want to use Python to construct a decision tree to fulfill this task. Please answer the following questions.

- Without using a decision tree, investigate the relation between the annual income and the other demographic predictors. Write a short summary of your observations. Which variables seem to be useful for predicting income? Why?
- Divide your data into 60% train and 40% test. Create the default C&R decision tree. How many leaves are in the tree? What is the size of each leaf node?
- What are the major predictors of income? How are they related to your observations in part (a)? How well your decision tree can predict a household's income in your training data?
- Give two decision rules that are useful for predicting income. Justify your choice.
- Using your decision tree, predict the annual income of the households in the test data. What is the error of your predictions?
- Describe the profile/characteristics of the households in your training data that are likely to have high income.
- The dataset *income_big.csv* consists of the marketing database with 10 spurious variables added. Each of the added variables takes integer values from 1 to 9 with equal probability (very often in data mining datasets there are predictors that are not related to the response variable). Run a C&R tree model using this data and compare your results with those of problem (c).
- Create two more C&R trees. Change the parameters of the tree such that you get a tree as large as possible (minimum-pruning). The other does prune with making sure you require at least 500 records in a parent nodes and 100 records in leaf nodes. How do the trees differ (briefly explain). Which seems more accurate on the training data? Which seems more accurate on the test data?
- In this part we want to check the effect of the size of training and testing data on the quality of the predictions. Construct a decision tree on different size of training data and compare the results. For example, consider a partition of the data into 50% for Training and 50% for Testing. What model performance do you obtain? Consider partitions of the data into 70% for Training and 30% for Testing, and 90% for Training and 10% for Testing and report on model and performance comparisons. Feel free to experiment with other size partitions on the data. Is there any specific model you would prefer to implement? Explain your reasoning.
- Construct two decision trees, once using "gini" measure once using "information gain" and try to keep the other parameters the same. Which model is better? In general, how do you choose the purity measure?
- Train a Random Forest classifier on the training data (using the same 60/40 split as in part b). Use an ensemble of 100 trees (`n_estimators=100`) and set `random_state=42` for reproducibility. What is the Out-of-Bag (OOB) error estimate of your model?
- Plot the feature importances as determined by the Random Forest model. How does the ranking of the most

important features compare to the major predictors you identified in the single decision tree (part c)? Provide a brief explanation for any similarities or differences.

- m. Predict the annual income of households in the test set using the Random Forest model. Calculate the prediction error and compare it to the error you obtained from the single decision tree in part (c). Which model performs better on the test set?
- n. Experiment with the number of trees in the forest. Train models with `n_estimators = [10, 50, 100, 200]` and plot the OOB error against the number of trees. At what point does the error seem to stabilize? What is the trade-off in using more trees?

